



Tsinghua University

Modern Computer Architecture Fall 2025

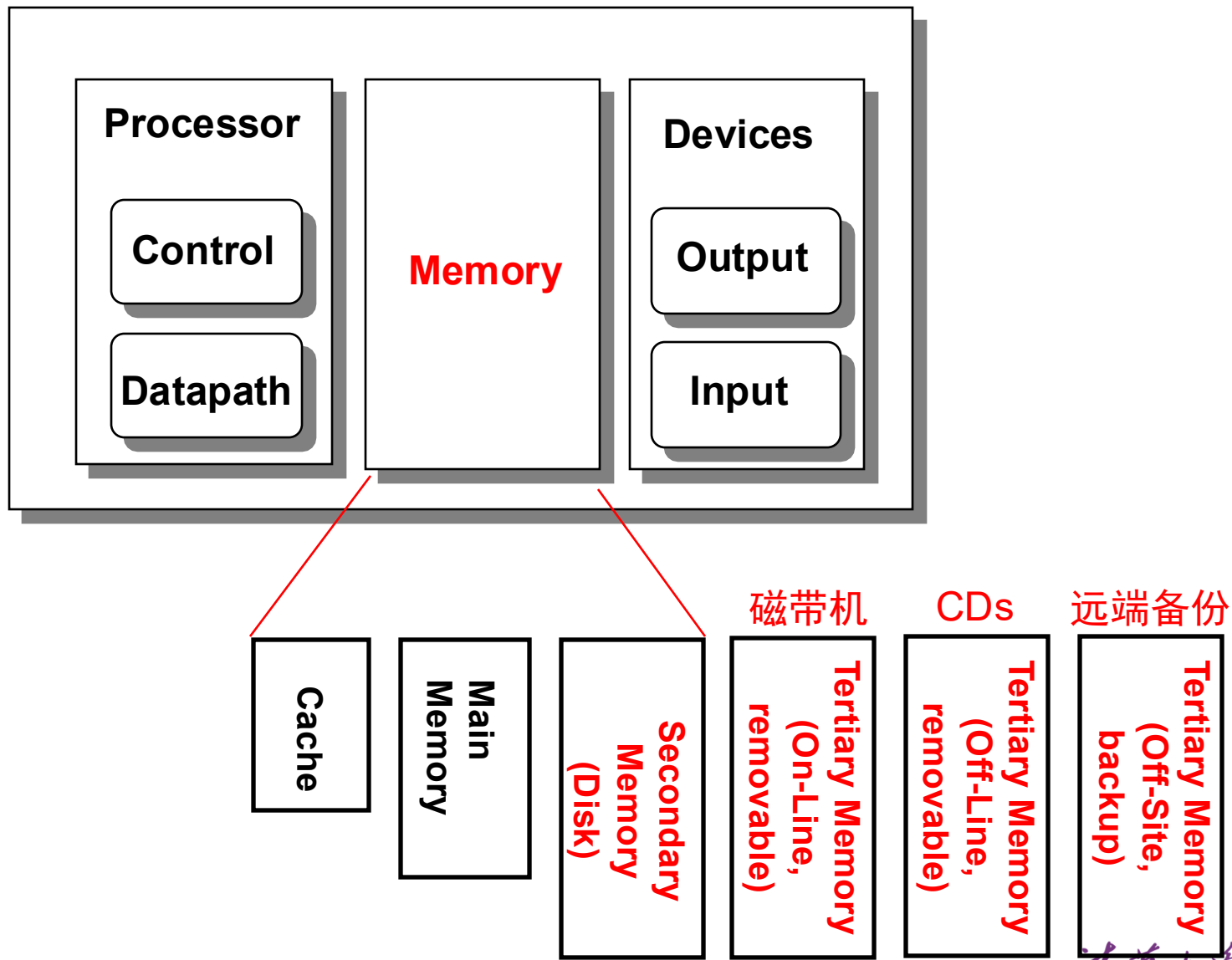
Lecture 09_1: Disk and RAIDs

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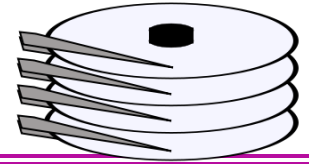
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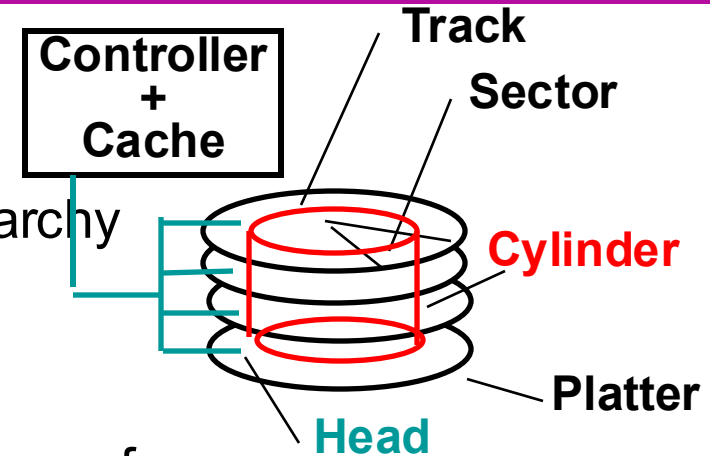
Review: Major Components of a Computer



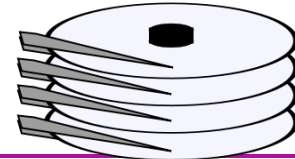
Magnetic Disk



- Purpose
 - Long term, **nonvolatile** storage
 - Almost lowest level in the memory hierarchy
 - slow, large, inexpensive
- General structure
 - A rotating platter coated with a magnetic surface
 - A moveable read/write head to access the information on the disk
- Typical numbers
 - 1 to 4 (1 or 2 surface) platters per disk of 1" to 5.25" in diameter (3.5" dominate in 2004)
 - Rotational speeds of 5,400 to 7200, 15,000 RPM
 - 10,000 to 50,000 **tracks** per surface
 - **cylinder** - all the tracks under the head at a given point on all surfaces
 - 100 to 500 **sectors** per track
 - the smallest unit that can be read/written (typically 512B)



Magnetic Disk Characteristic



- Disk read/write components

1. **Seek time**: position the head over the proper track (3 to 13 ms avg)

- due to locality of disk references the actual average seek time may be only 25% to 33% of the advertised number

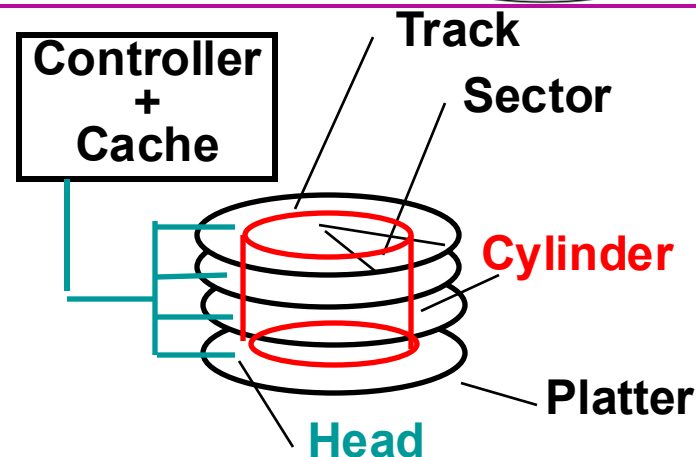
2. **Rotational latency**: wait for the desired sector to rotate under the head ($\frac{1}{2}$ of $1/\text{RPM}$ converted to ms)

- $0.5/5400\text{RPM} = 5.6\text{ms}$ to $0.5/15000\text{RPM} = 2.0\text{ms}$

3. **Transfer time**: transfer a block of bits (one or more sectors) under the head to the disk controller's cache (70 to 125 MB/s @ 2008 are typical disk transfer rates)

- the disk controller's "cache" takes advantage of spatial locality in disk accesses
 - cache transfer rates are much faster (e.g., 375 MB/s)

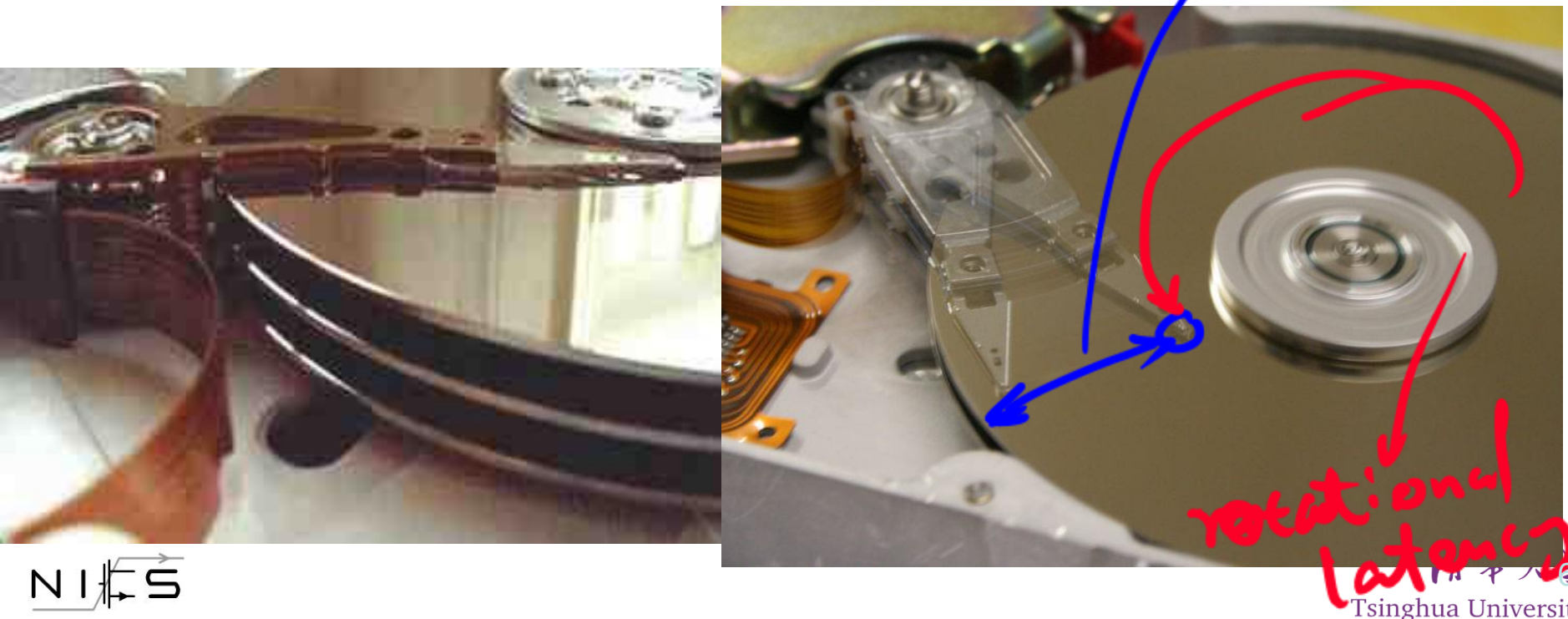
4. **Controller time**: the overhead the disk controller imposes in performing a disk I/O access (typically $< .2\text{ ms}$)





Aside: Real-world Data

- **BlackFriday!**
 - Seagate 250GB Internal Hard Drive - \$39.99
 - PNY 512MB PC2700 DDR Memory - \$49.99
- **Manufacturers:**
 - 98% of the market are from Seagate, Western Digital, Samsung, Hitachi, Toshiba



Typical Disk Access Time

- The average time to read or write a 512B sector for a disk rotating at 10,000RPM with average seek time of 6ms, a 50MB/sec transfer rate, and a 0.2ms controller overhead

If the measured average seek time is 25% of the advertised average seek time, then

Typical Disk Access Time

- The average time to read or write a 512B sector for a disk rotating at 10,000RPM with average seek time of 6ms, a 50MB/sec transfer rate, and a 0.2ms controller overhead

$$\text{Avg disk read/write} = 6.0\text{ms} + 0.5 / (10000\text{RPM} / (60\text{sec/minute})) + 0.5\text{KB} / (50\text{MB/sec}) + 0.2\text{ms} = 6.0 + 3.0 + 0.01 + 0.2 = 9.21\text{ms}$$

If the measured average seek time is 25% of the advertised average seek time, then

$$\text{Avg disk read/write} = 1.5 + 3.0 + 0.01 + 0.2 = 4.71\text{ms}$$

- The rotational latency is usually the largest component of the access time

Disk Interface Standards

- Higher-level disk interfaces have a microprocessor disk controller that can lead to performance optimizations
 - **ATA** (Advanced Technology Attachment) – An interface standard for the connection of storage devices such as hard disks, solid-state drives, and CD-ROM drives. **Parallel ATA has been largely replaced by Serial ATA.**
 - **SCSI** (Small Computer Systems Interface) – A set of standards (commands, protocols, and electrical and optical interfaces) for physically connecting and transferring data between computers and peripheral devices. Most commonly used for hard disks and tape drives. **SAS = Serial Attached SCSI** is the newer version.
- In particular, disk controllers have **SRAM disk caches** which support fast access to data that was recently read and often also include **prefetch** algorithms to try to anticipate demand

Magnetic Disk Examples (www.seagate.com)

Characteristic	Seagate ST37	Seagate ST32	Seagate ST94
Disk diameter (inches)	3.5	3.5	2.5
Capacity (GB)	73.4	200	40
# of surfaces (heads)	8	4	2
Rotation speed (RPM)	15,000	7,200	5,400
Transfer rate (MB/sec)	57-86	32-58	34
Minimum seek (ms)	0.2r-0.4w	1.0r-1.2w	1.5r-2.0w
Average seek (ms)	3.6r-3.9w	8.5r-9.5w	12r-14w
MTTF (hours@25°C)	1,200,000	600,000	330,000
Dimensions (inches)	1"x4"x5.8"	1"x4"x5.8"	0.4"x2.7"x3.9"
GB/cu.inch	3	9	10
Power: op/idle/sb (watts)	20/12/-	12/8/1	2.4/1/0.4
GB/watt	4	16	17
Weight (pounds)	1.9	1.4	0.2

Magnetic Disk Examples (www.seagate.com)

Feature	Seagate ST31000340NS	Seagate ST973451SS	Seagate ST9160821AS
Disk diameter (inches)	3.5	2.5	2.5
Capacity (GB)	1000	73	160
# of surfaces (heads)	4	2	2
Rotation speed (RPM)	7,200	15,000	5,400
Transfer rate (MB/sec)	105	79-112	44
Minimum seek (ms)	0.8r-1.0w	0.2r-0.4w	1.5r-2.0w
Average seek (ms)	8.5r-9.5w	2.9r-3.3w	12.5r-13.0w
MTTF (hours@25°C)	1,200,000	1,600,000	??
Dimensions (inches), Weight (lbs)	1x4x5.8, 1.4	0.6x2.8x3.9, 0.5	0.4x2.8x3.9, 0.2
GB/cu.inch, GB/watt	43, 91	11, 9	37, 84
Power: op/idle/sb (watts)	11/8/1	8/5.8/-	1.9/0.6/0.2
Price in \$/GB	~\$0.3/GB	~\$5/GB	~\$0.6/GB

Disk Latency & Bandwidth Milestones

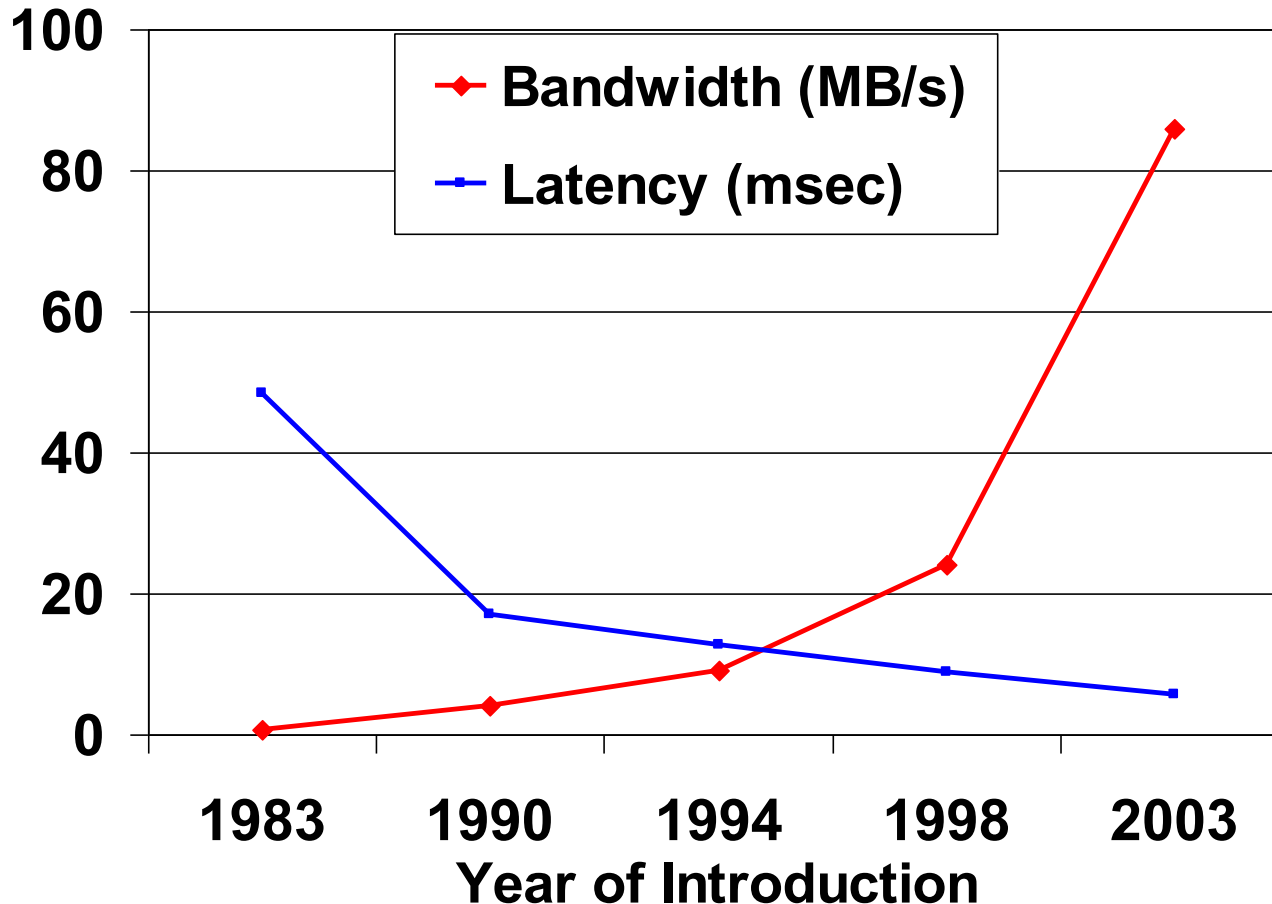
	CDC Wren	SG ST41	SG ST15	SG ST39	SG ST37
RSpeed (RPM)	3600	5400	7200	10000	15000
Year	1983	1990	1994	1998	2003
Capacity (Gbytes)	0.03	1.4	4.3	9.1	73.4
Diameter (inches)	5.25	5.25	3.5	3.0	2.5
Interface	ST-412	SCSI	SCSI	SCSI	SCSI
Bandwidth (MB/s)	0.6	4	9	24	86
Latency (msec)	48.3	17.1	12.7	8.8	5.7

Patterson, CACM Vol 47, #10, 2004

- Disk latency
 - average seek time + rotational latency.
- Disk bandwidth is the peak transfer time of formatted data from the media.

Latency & Bandwidth Improvements

- In the time that the disk **bandwidth doubles** the **latency** improves by a factor of only 1.2 to 1.4



Aside: Media Bandwidth/Latency Demands

- Bandwidth requirements
 - High quality video
 - Digital data = $(30 \text{ frames/s}) \times (640 \times 480 \text{ pixels}) \times (24\text{-b color/pixel}) = 221 \text{ Mb/s} \text{ } (27.625 \text{ MB/s})$
 - High quality audio
 - Digital data = $(44,100 \text{ audio samples/s}) \times (16\text{-b audio samples}) \times (2 \text{ audio channels for stereo}) = 1.4 \text{ Mb/s} \text{ } (0.175 \text{ MB/s})$
 - **Compression** *reduces* the bandwidth requirements considerably
- Latency issues
 - How sensitive is your eye (ear) to variations in video (audio) rates?
 - How can you ensure a constant rate of delivery?
 - How important is synchronizing the audio and video streams?
 - 15 to 20 ms early to 30 to 40 ms late is tolerable

Flash Storage - SSD

- Flash memory is the first credible challenger to disks. It is **semiconductor** memory that is nonvolatile like disks, but has latency 100 to 1000 times faster than disk and is smaller, more power efficient, and more shock resistant.
 - In 2008, the price of flash was **\$4 to \$10 per GB or about 2 to 40** times higher than disk and 5 to 10 times lower than DRAM.
 - Flash memory bits **wear out** (unlike disks and DRAMs), but **wear leveling** can make it unlikely that the write limits of the flash will be exceeded.
 - Benefits of Moore's Law

Feature	Kingston	Transend	RiDATA
Capacity (GB)	8	16	32
Bytes/sector	512	512	512
Transfer rates (MB/sec)	4	20r-18w	68r-50w
MTTF (hrs)	>1,000,000	>1,000,000	>4,000,000
Price (2008)	~ \$30	~ \$70	~ \$300

Flash Storage – an Example

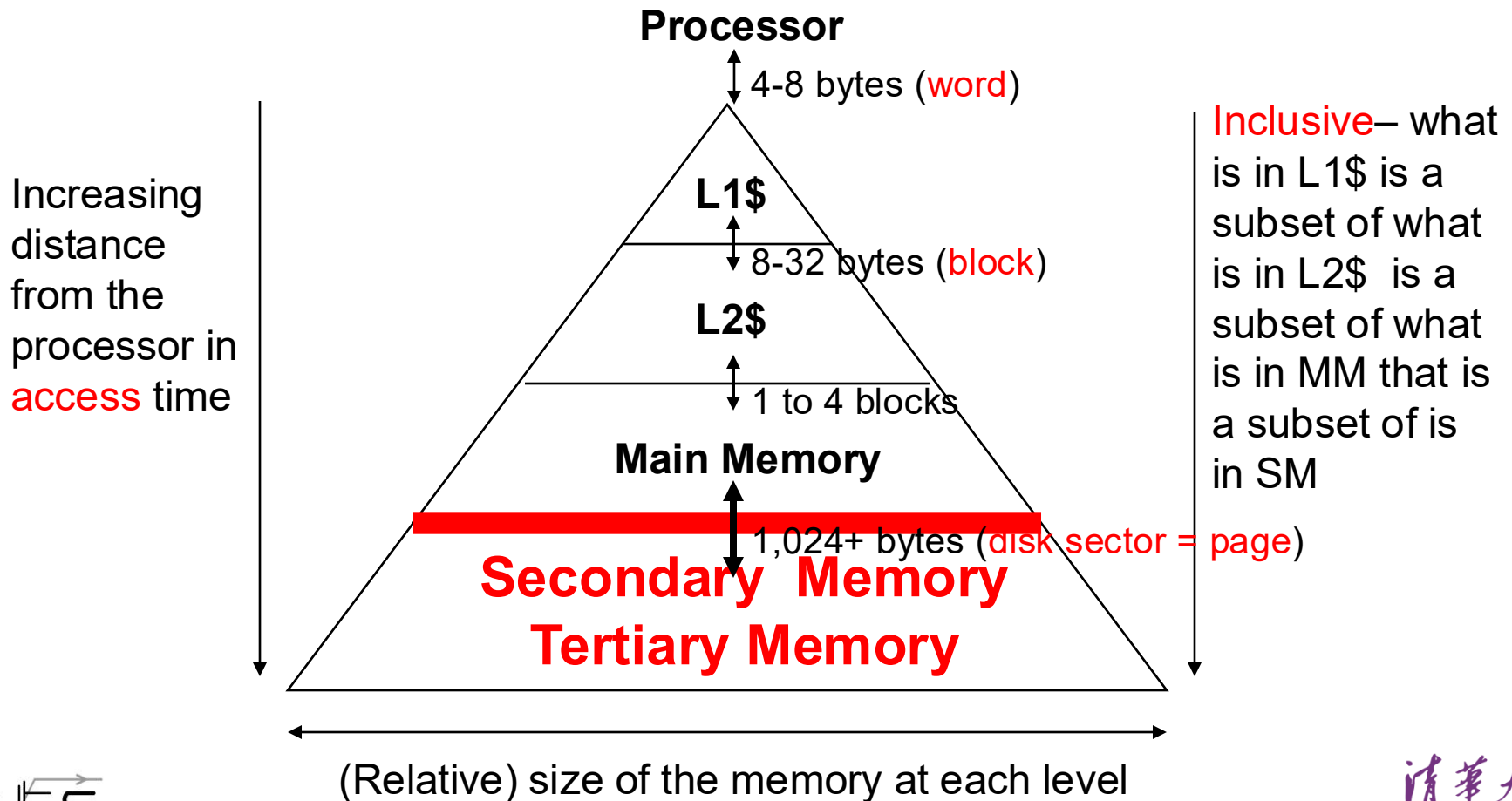
- Flash memory is organized into blocks of pages
 - page size 512 B – 4 KB, block size 32 – 128 pages (typical)
 - “read” is page read
 - “write” is block erase (~ 1 ms) followed by page write (and additional copying of other pages in that block)
- Intel X25-E SATA solid state drive

sequential read throughput	250 MB/s
random read throughput	140 MB/s
random read access time	30 microsec

sequential write throughput	170 MB/s
random write throughput	14 MB/s
random write access time	300 microsec

Review: The Memory Hierarchy

- Take advantage of the principle of locality to present the user with **as much memory as is available** in the **cheapest** technology at the speed offered by the **fastest** technology

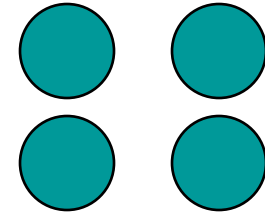


Dependability, Reliability, Availability

- Reliability is measured by the **mean time to failure** (MTTF). Service interruption is measured by **mean time to repair** (MTTR).
- Availability – a measure of service accomplishment
$$\text{Availability} = \text{MTTF} / (\text{MTTF} + \text{MTTR})$$
- To increase MTTF, either improve the quality of the components or **design the system to continue operating in the presence of faulty components**
 - Fault avoidance: preventing fault occurrence by construction
 - Level Holder with 2 inverters
 - **Fault tolerance**: using redundancy to correct or bypass faulty components (hardware faulty)
 - **Fault forecasting**: forecasting and replace before error

RAID: Disk Arrays

Redundant Array of Inexpensive Disks



- Arrays of small inexpensive disks, potential reduced **cost**
 - compared to a single large expensive disk with similar capacity
- Potential increased **throughput** by using many disk drives
 - Data can be spread out across multiple disks
 - Multiple disk accesses can be made simultaneously
- But, **reliability** for the array is **lower** than for a single disk
- **Availability** can be improved by adding redundant disks
 - Lost information can be reconstructed from redundant information
 - MTTR: mean time to repair is measured in **hours**
 - MTTF: mean time to failure of disks is measured in **many-years**

RAID: Design Choices

- What are we trying to achieve?
 - increased storage capacity --
 - increased reliability / availability --
 - increased throughput --
 - for a single large request
 - for many independent small requests
 - How much extra will it cost?
 - RAID controller in front of disk controllers
 - synchronization between the disks?
 - additional disks?
- Methods
- multiplicity of disks
 - mirroring, parity, ECC
 - striping, distribution

RAID: Design Choices

- Performance vs. Reliability / Availability
 - Level 0 – striping
 - Level 1 – mirroring (usually also with striping)
 - Level 2 – bit-wise striping with ECC across all disks
 - Level 3 – bit-wise striping with parity across groups of disks
 - Level 4 – striping with parity blocks
 - Level 5 – striping with distributed parity blocks
 - Level 6 – striping with additional distributed parity blocks
- 3个共同特性
 - RAID是一组物理磁盘驱动器，在OS下被视为一个单一的逻辑驱动器
 - 数据分布在一组物理磁盘上
 - 冗余磁盘容量用于存储奇偶校验信息，用于数据恢复

Patterson et al., 1988/89, UC Berkeley (levels 0-6)

RAID: Data Mapping, Blocks, Strips, Stripes

- A RAID system consists of multiple small **physical** disks but is treated as one large **logical** disk.
 - The **logical disk sectors** are L_k , $k = 0, 1, 2, \dots$ (consecutive).
 - The **physical disk sectors** are DiS_j (disk i , sector j).
 - We need a **mapping** from k to (i,j) . **There are many choices.**
- A **block** is some fixed-size unit of data on the disks – usually some number of consecutive sectors.
 - For ex., the operating system and file system might prefer to deal with pages or clusters of pages.
 - Think in terms of **logical blocks**, not in terms of physical sectors.
- A **strip** is some fixed-size number of consecutive blocks.
 - **Striping** is the mapping of strip k to disk $(k \bmod N)$ when we are using N disks.
 - A **stripe** is a set of N consecutive strips that has mapped to N separate disks.

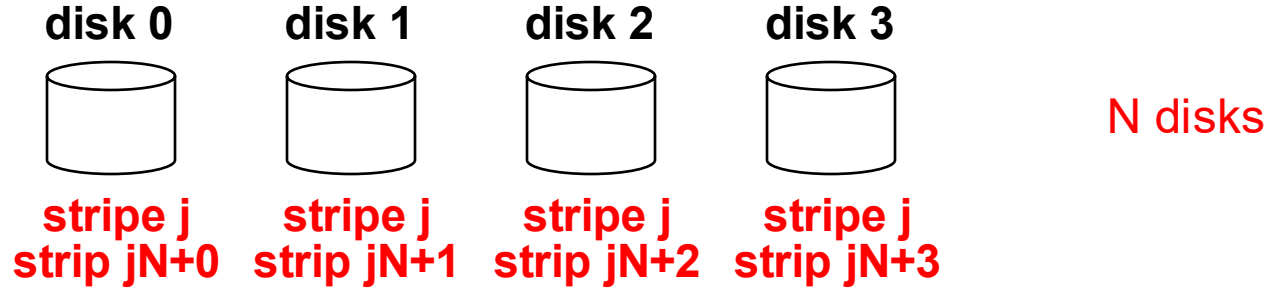
RAID: Design Parameters

- Striping – higher data rate for large requests
- Distribution – higher data rate for multiple concurrent requests
- Mirroring, Parity – **higher availability at various costs**

Category	Level	Disks	Features
Striping	0	N	No fault tolerance
Mirroring	1	2N	Expensive
Parallel access, Parity disk(s)	2	$N + m(<N)$	Hamming code
	3	$N + 1$	Parity only
Independent access, Parity blocks	4	$N + 1$	Parity only, on one disk
	5	$N + 1$	Distributed parity
	6	$N + 2$	Dual distributed parity

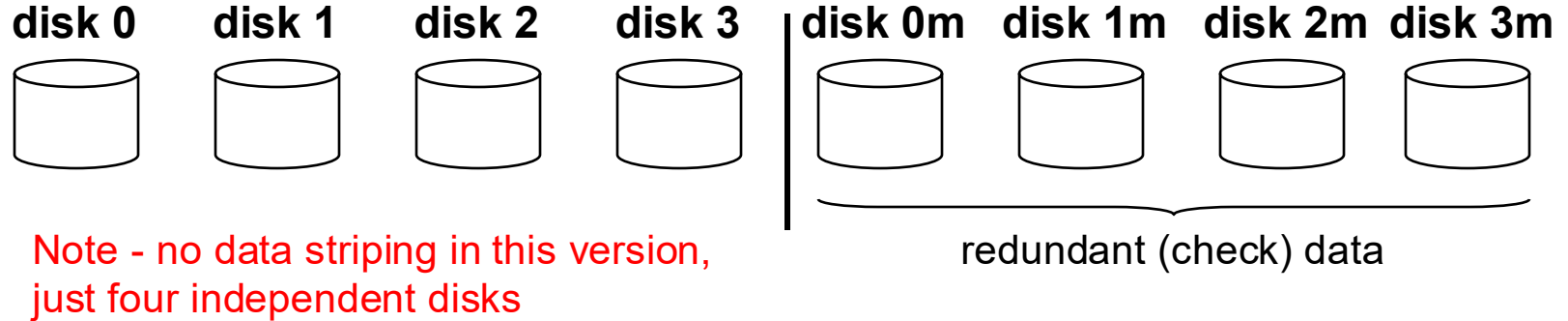
N data disks; m proportional to $\log N$

RAID: Level 0 (No Redundancy; Striping)



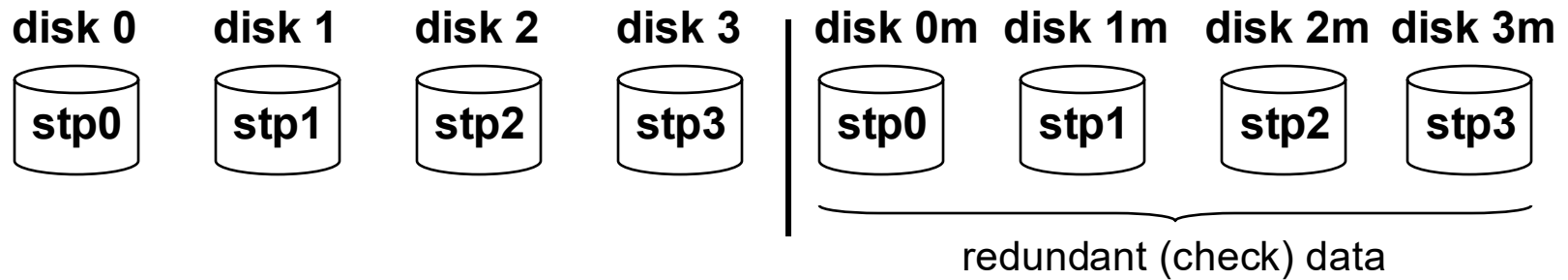
- Multiple smaller disks as opposed to one big disk
 - Spreading the data over multiple disks – **striping** – means that multiple blocks can be accessed in parallel increasing the performance – this works well for **large data requests**
 - A 4 disk system gives four times the throughput of a 1 disk system, assuming equivalent I/O channels on each disk
 - This **does not work** well for **small requests at random locations** – usually only one disk is active
 - Same cost as one *big* disk – assuming 4 small disks (plus a new controller) cost the same as one big disk
- No redundancy, so what if one disk fails?
 - Failure of one or more disks is more likely as the number of disks in the system increases

RAID: Level 1 (Redundancy via Mirroring)



- Uses **twice** as many disks as RAID 0 (e.g., 8 smaller disks with the second set of 4 duplicating the first set) so there are always **two copies of the data**
 - # redundant disks = # of data disks, so twice the cost of one big disk
 - writes have to be made to both sets of disks, so writes would be only 1/2 the performance of an equivalent 8-disk RAID 0
 - reads are at full rate, and can **use either or both of two disks**
- What if one (known) disk fails?
 - If a disk fails, the system just goes to the “**mirror**” for the data

RAID: Level 0+1 (Striping with Mirroring)



- Combines the best of RAID 0 and RAID 1, data is striped across four disks and mirrored to four disks
 - **Four times** the throughput (due to striping)
 - # redundant disks = # of data disks, so twice the cost of one big disk
 - writes have to be made to both sets of disks, so writes would be only 1/2 the performance of an equivalent 8-disk RAID 0
 - reads are at full rate, can **use either or both sets of disks**
- What if one (known) disk fails?
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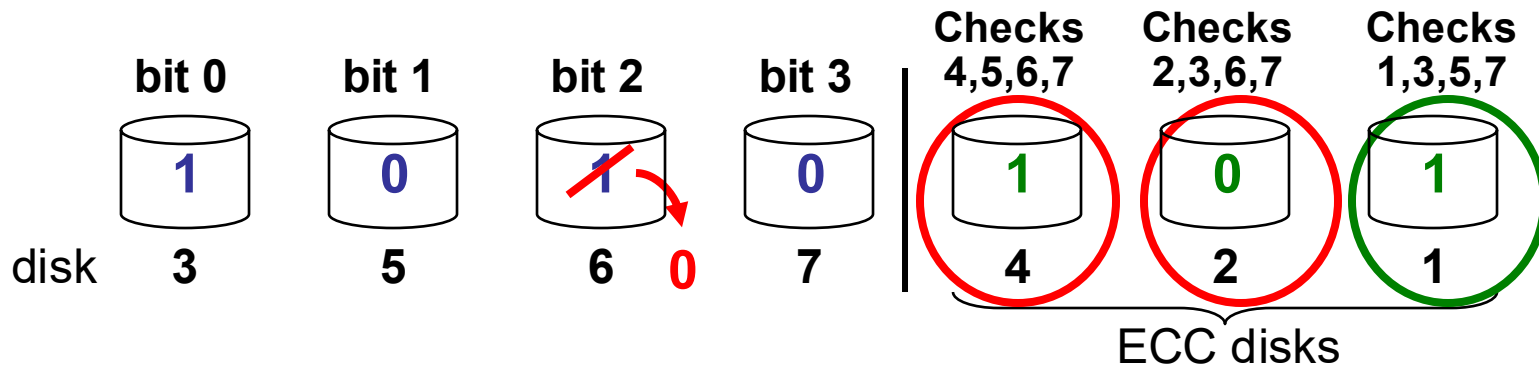
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	6	N + 2	Dual distributed parity

N data disks; m proportional to log N

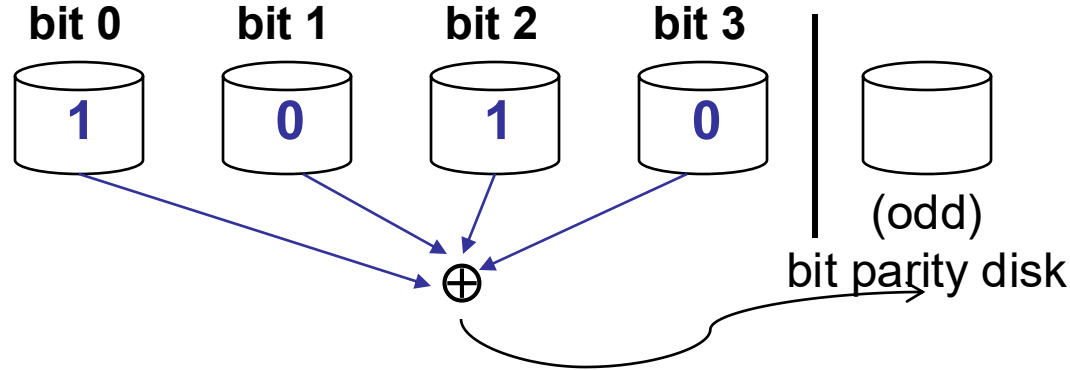
RAID: Level 2 (Redundancy via ECC)



ECC disks 4 and 2 point to either data disk 6 or 7, but ECC disk 1 says disk 7 is okay, so disk 6 must be in error

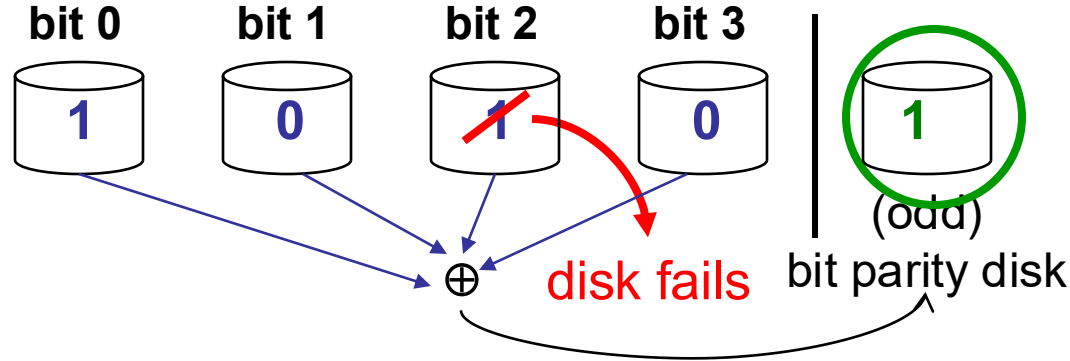
- ECC disks contain the parity of data on a set of **distinct overlapping** disks
 - # redundant disks = by **Hamming Rule**, so almost twice the cost of one big disk
 - writes require **computing parity** to **write to the ECC disks**
 - reads require reading ECC disks and confirming parity
- Can tolerate *limited* (single) disk failure, since the data can be reconstructed
 - use smaller units of data than a strip, tightly synchronize the disks

RAID: Level 3 (Bit-Interleaved Parity)



- N is the number of disks in a **protection group**
 - # redundant disks = 1 × # of protection groups
 - writes require writing the new data to the data disk as well as computing the parity, meaning reading the other disks, so that the parity disk can be updated
 - reads require reading all the operational data disks as well as the parity disk to calculate the missing data that was stored on the failed disk
- Can tolerate *limited* (single known) disk failure, since the data can be reconstructed

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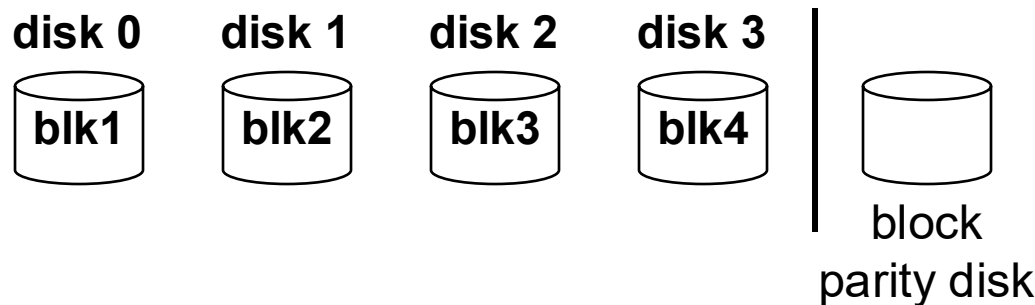
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N data disks; m proportional to log N

RAID: Level 4 (Block-Interleaved Parity)

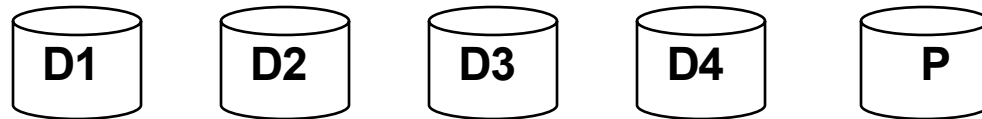


- The parity is stored as **blocks** associated with sets of data blocks
 - # redundant disks = $1 \times \#$ of protection groups
 - Supports “**small reads**” and “**small writes**” (reads and writes that go to just one (or a few) data disks in a protection group)
 - by watching which bits change when writing new information, only need to change the corresponding bits on the parity disk
 - the parity disk must be updated on every write, so it is a bottleneck while **Independent access to data disk**
- Can tolerate *limited* (single known) disk failure, since the data can be reconstructed
 - Tight synchronization of the disks is not required.

Small Writes

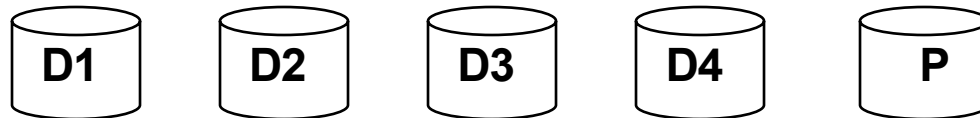
- RAID 3 writes – to one data disk only, update parity disk

New D1 data



- RAID 4 writes – to one data disk only, update parity disk

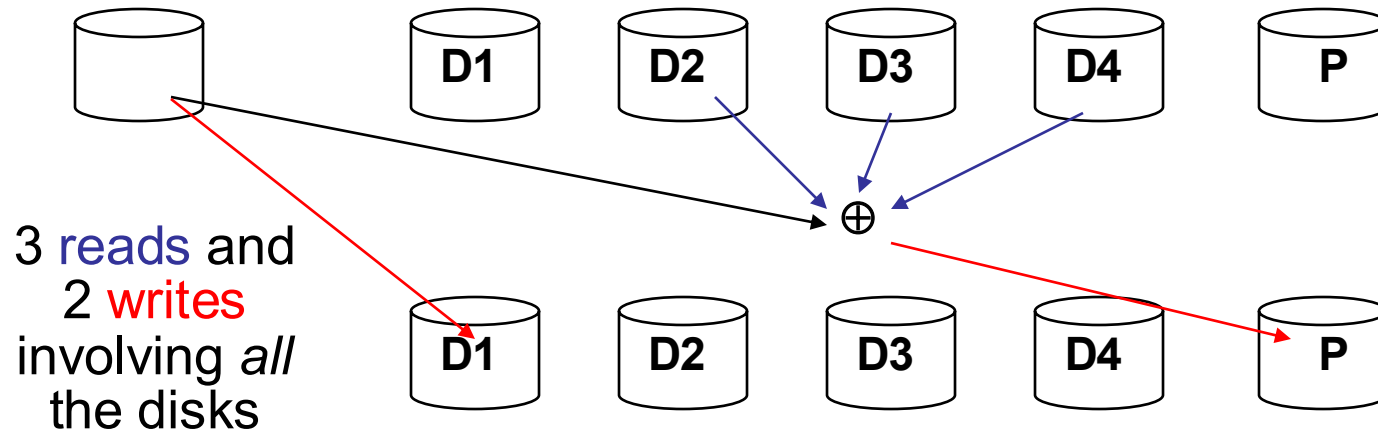
New D1 data



Small Writes

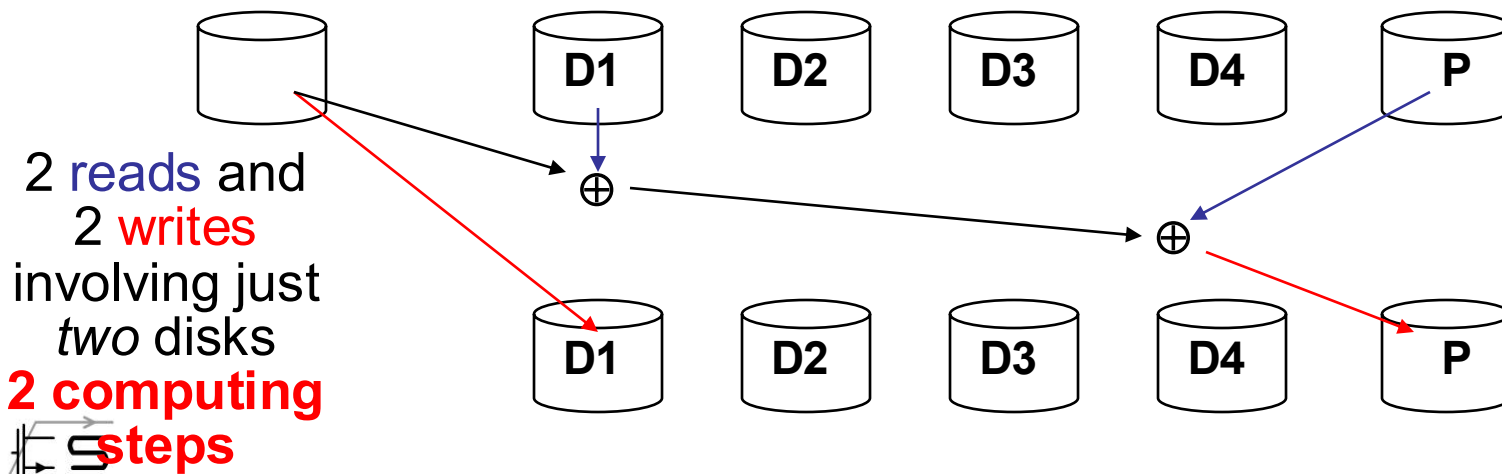
- RAID 3 writes – to one data disk only, update parity disk

New D1 data

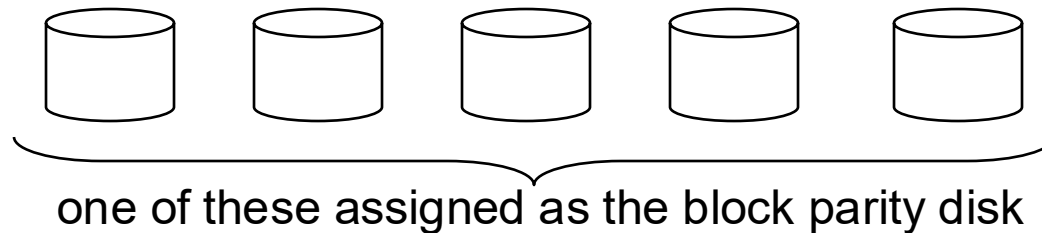


- RAID 4 writes – to one data disk only, update parity disk

New D1 data



Level 5 (**Distributed** Block-Interleaved Parity)

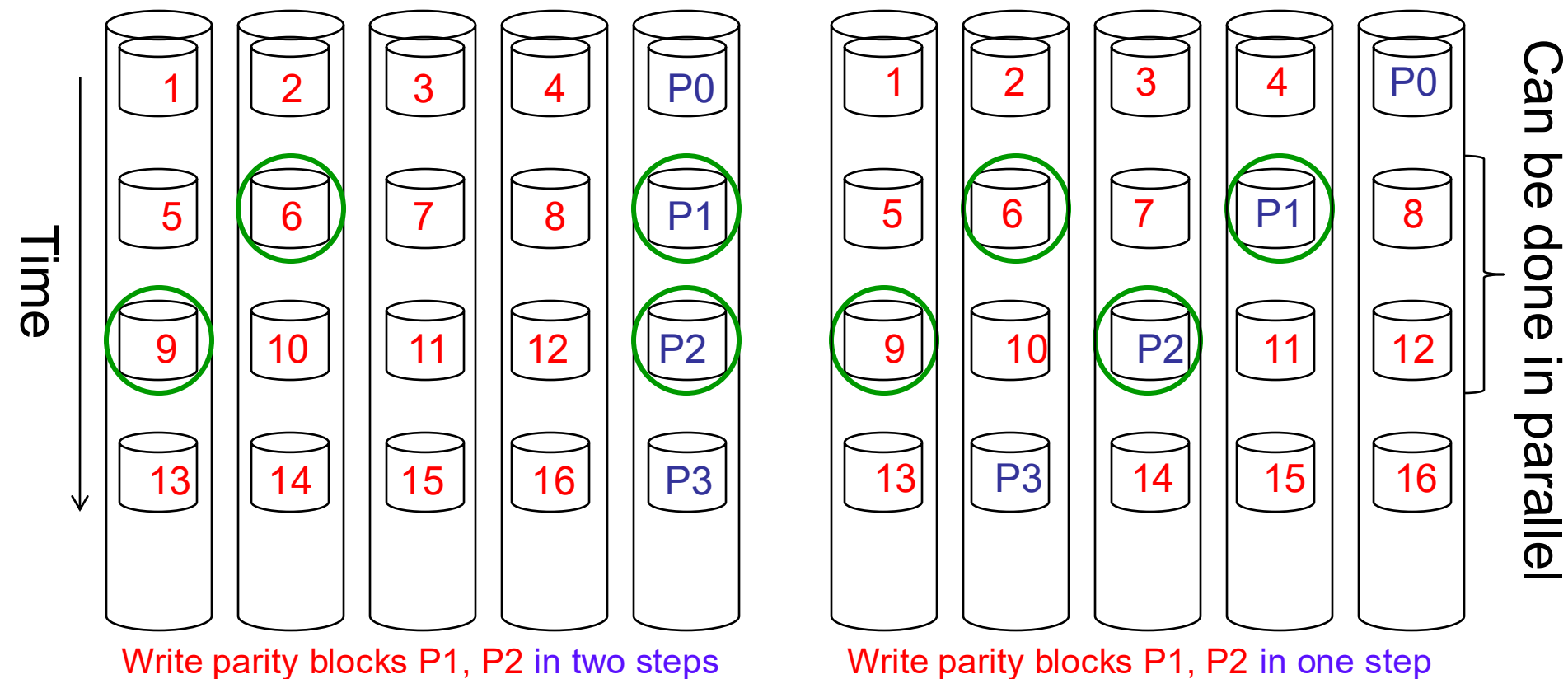


- the parity block can be located on any of the disks so there is no single bottleneck for writes
 - # redundant disks = $1 \times \#$ of protection groups
 - Supports “small reads” and “small writes” (reads and writes that go to just one (or a few) data disks in a protection group)
 - Allows multiple simultaneous writes as long as the accompanying parity blocks are not located on the same disk
- Can tolerate *limited* (single known) disk failure, since the data can be reconstructed

Distributing Parity Blocks

RAID 4

RAID 5



- By distributing parity blocks to all disks, some small writes can be performed in parallel (BIK 5 & 10)

RAID: Design Parameters

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- Distribution – higher data rate for multiple concurrent requests
- Mirroring, Parity – higher availability at various costs

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N data disks; m proportional to log N

RAID: Summary of Goals and Problems

- Failure-resistant
 - protect against **data loss** due to disk failure
- Failure-tolerant
 - protect against **loss of data access** due to failure of any single component
- Disaster-tolerant
 - two or more independent sets of disks, either of which provides access to stored data

-
- Problem areas
 - correlated failures (error correction assumes independent errors)
 - atomic updates in database transactions
 - timing issues – each drive does its own error correction, etc.
 - controller issues – data portability, single point of failure, etc.

Summary

- Four components of disk access time:
 - Seek Time: advertised to be 3 to 13 ms but lower in real systems
 - Rotational Latency: from 5.6 ms at 5400 RPM to 2.0 ms at 15000 RPM (average times)
 - Transfer Time: 70 to 125 MB/s for contiguous sectors
 - Controller Time: typically less than .2 ms
- RAIDs can be used to improve availability
 - RAID 1, 4, 5 – widely used in servers, one estimate is that 80% of disks in servers are RAIDs
 - Lots of suppliers – EMC, IBM, HP, etc.

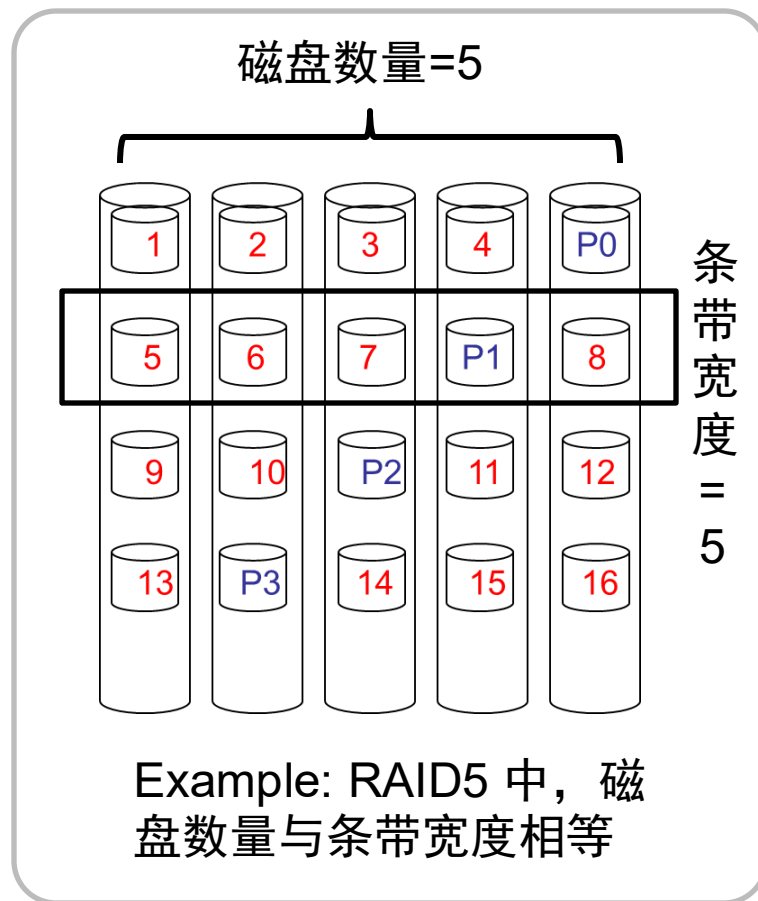
Recent Research on RAID

- 传统 RAID 方案缺点：条带宽度 = 磁盘数量

- 单盘失效时：健康盘全部读一遍、热备盘全部写一遍
- 以 16 TB HDD，读写 200MB/s 为例，需要 22 小时
- 数据恢复过程中系统性能下降，且极易发生进一步的数据丢失

- RAID+

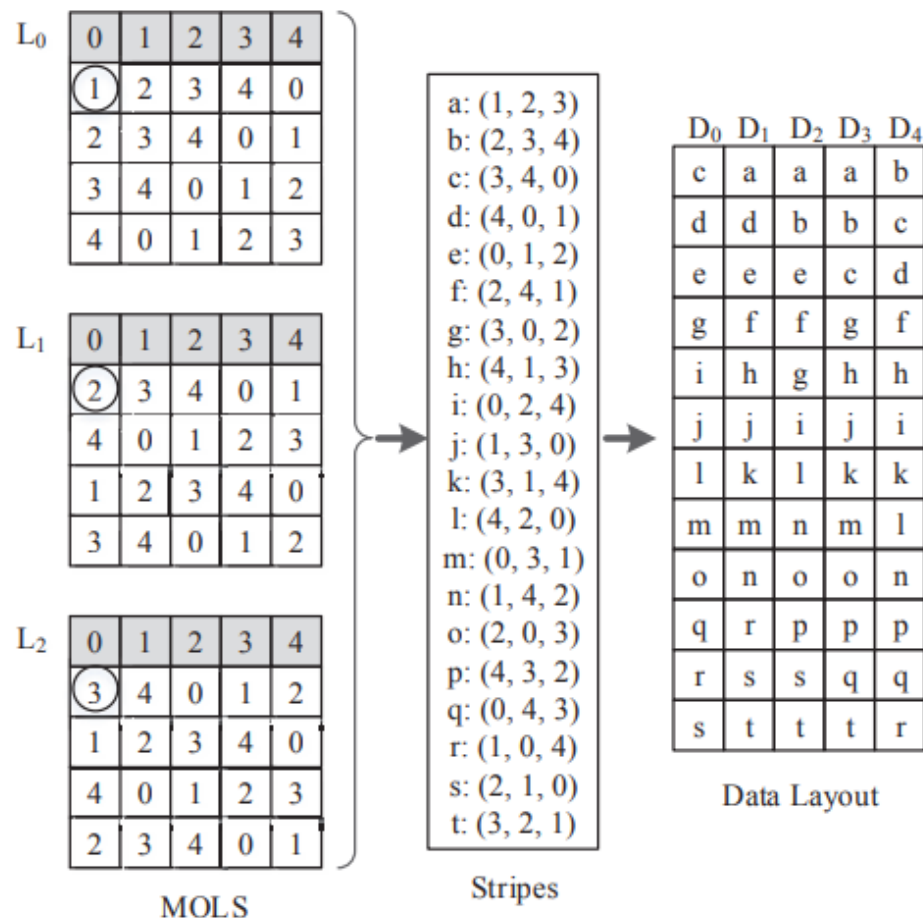
- 条带宽度 $\neq$ 磁盘数量



Zhang, Guangyan, et al. "RAID+: Deterministic and Balanced Data Distribution for Large Disk Enclosures." *16th USENIX Conference on File and Storage Technologies (FAST 18)*. 2018.

RAID+

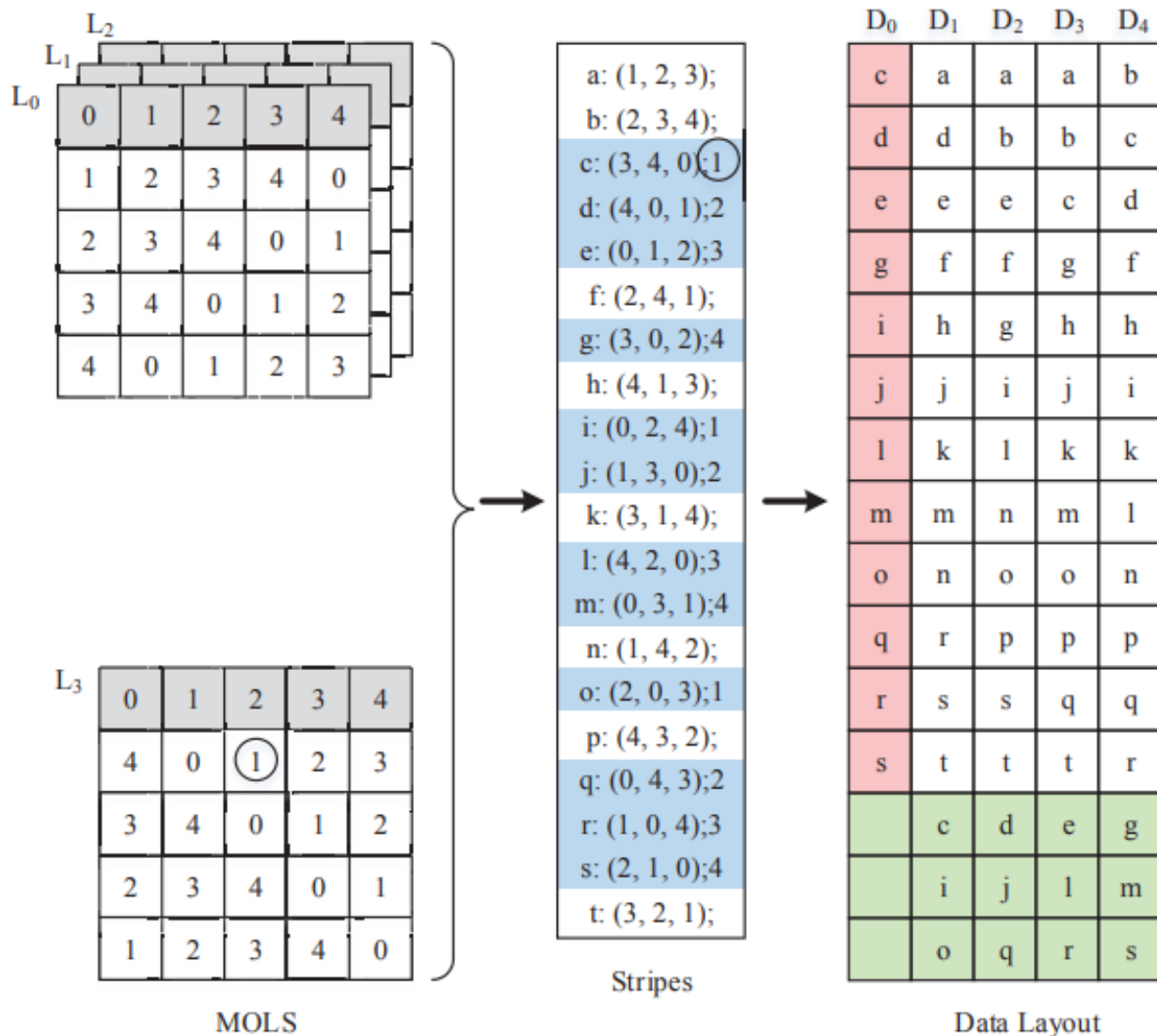
- 基于正交拉丁方 (Mutually Orthogonal Latin Squares, MOLS) 的数据分布设计
- 数学上可证明, 边长为 n 时, 最多可写出 $(n-1)$ 个正交拉丁方
- 因此对于 n 个硬盘, 可支持 $2 \sim n-1$ 的条带宽度



Example: 条带宽度=3; 磁盘数量=5

RAID+: 单盘失效时

- 取一个未使用的拉丁方，重新构建一个磁盘数量-1，但条带宽度不变的阵列
- 恢复过程中，所有读写都分散在阵列的每个硬盘上



RAID+: 性能提升

# of disks	RAID-50	RAID _R	RAID _H	RAID+
56 + 3	307s	60s	102s	41s
28 + 3	307s	99s	143s	83s

Table 2: Offline rebuild time comparison

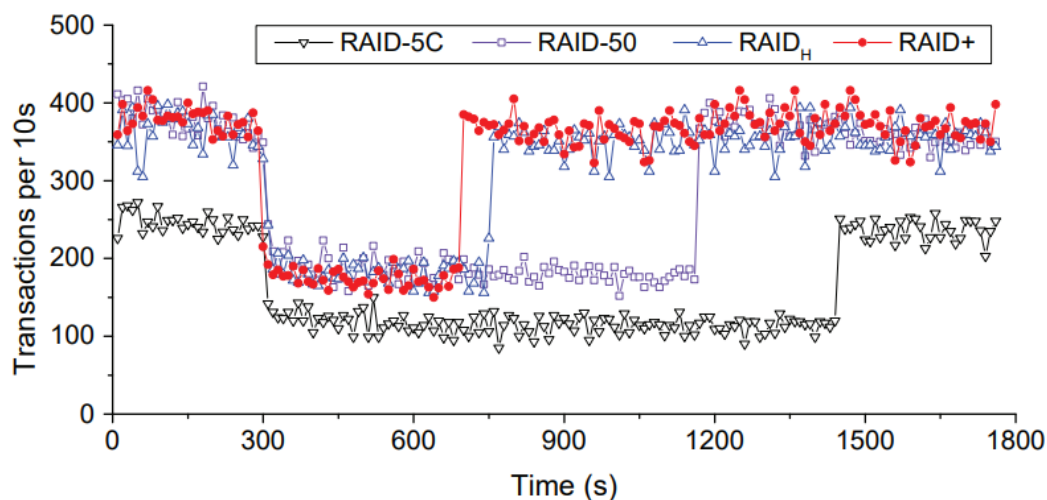


Figure 5: TPC-C online rebuild w. single-disk failure

Reading

- Computer Organization and Design: The Hardware / Software Interface, 4th
- 5.4
- 5.5
- 6.1 ~ 6.4
- 6.9